Tech Interview Prep:

Tool Use Mechanics, Function Calling Protocols & The Client-Model Interaction Loop

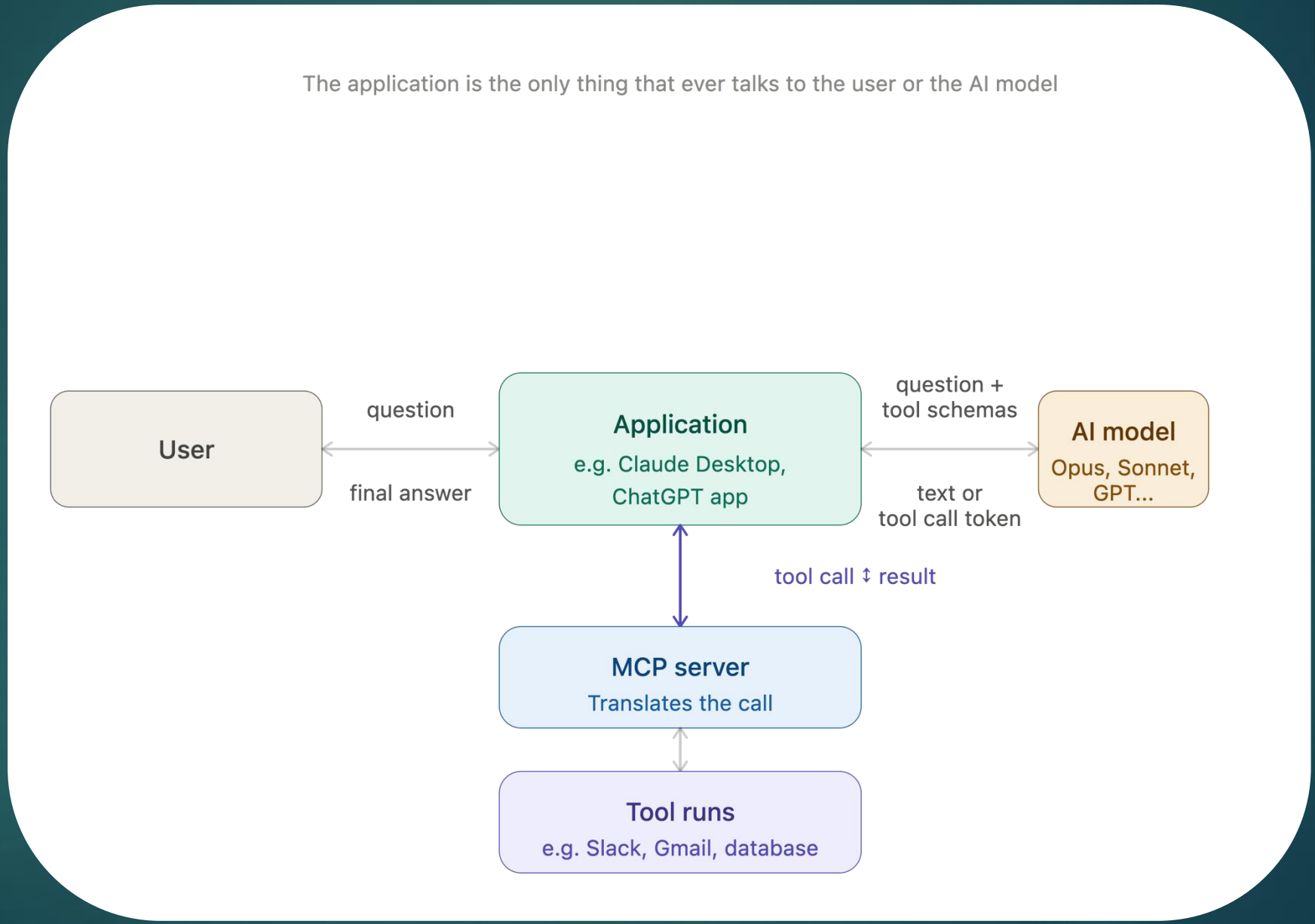
“How does an LLM know when and how to call an external tool? Describe the client-model interaction loop during function calling.”

Slide 1: Title & The Architectural Illusion

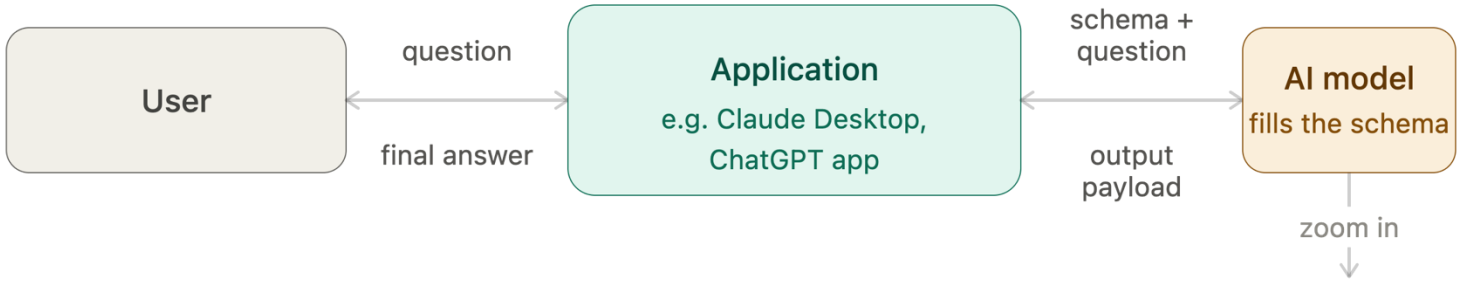
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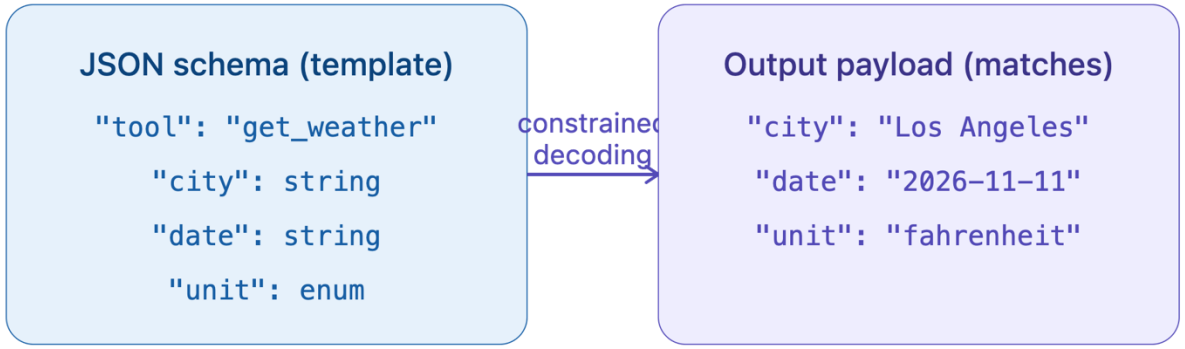
Slide 2: The “When”: Intent Detection & Tool Selection Mechanics



Slide 3: The “How”: Argument Generation & Schema Conformance



Inside the AI model: parameter extraction

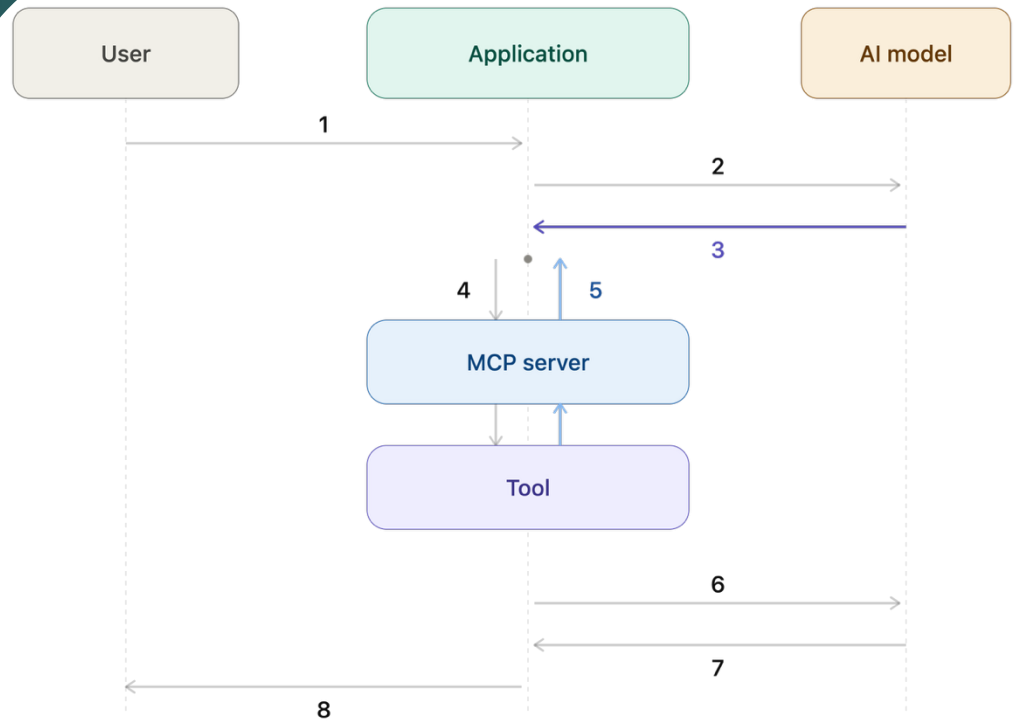


Type safety: numbers stay numbers, enums stay valid values

Missing a required field? The AI asks the user instead of guessing

The payload then travels back up: AI model → application → tool

Slide 4: The Client-Model Interaction Loop: End-To-End Walkthrough



Legend

1 — User → Application: sends the question

2 — Application → AI model: forwards question + tool schemas

3 — AI model → Application: "I need a tool" (tool_calls)

4 — Application → MCP server: sends the tool call

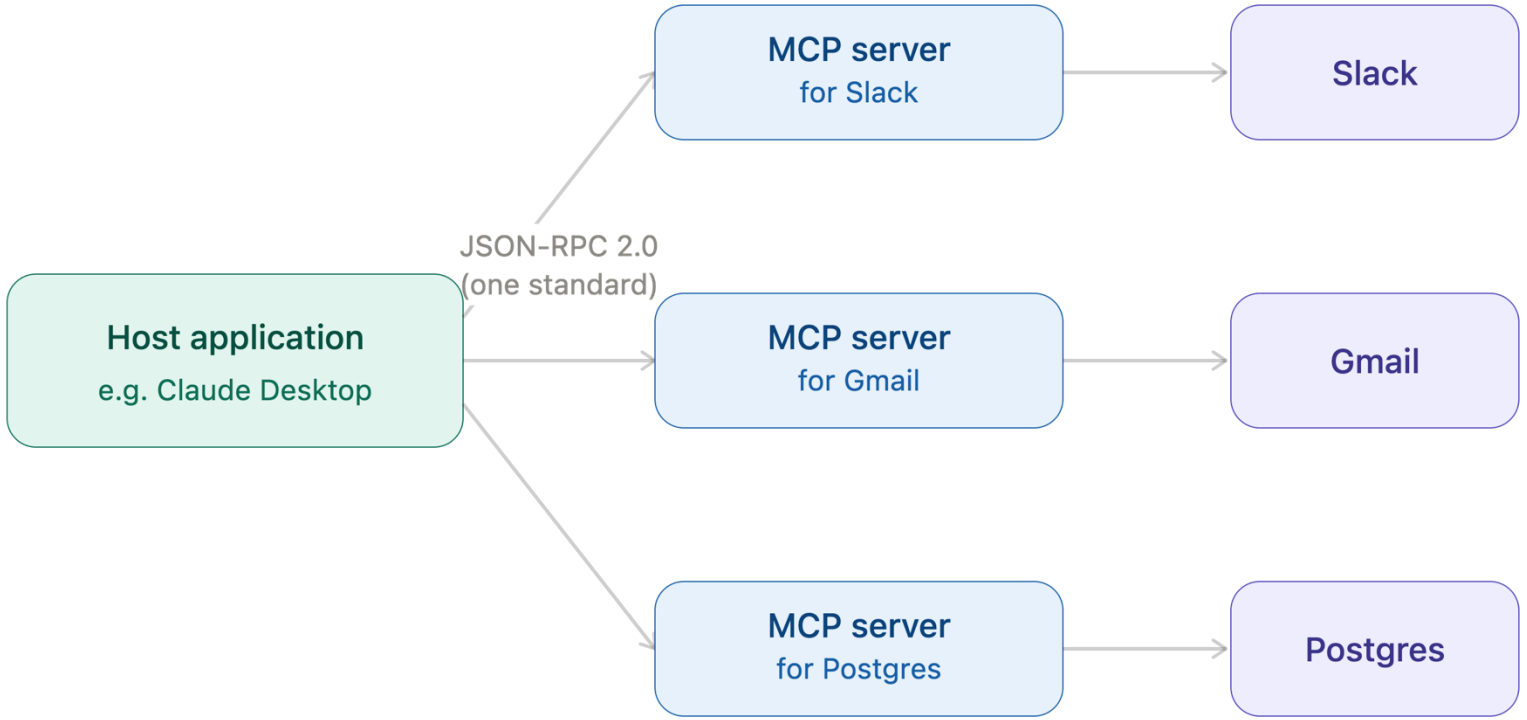
5 — MCP server → Application: returns the tool's result

6 — Application → AI model: re-injects the result

7 — AI model → Application: returns the final answer

8 — Application → User: delivers the final answer

Slide 5: Standardizing Tool Protocols: Model Context Protocol (MCP)



Before MCP

Every app built its own custom connection to every tool

— e.g. 5 apps × 5 tools = 25 custom integrations

After MCP

One MCP server per tool, one standard protocol — plug in once, works everywhere

Summary & Systems Engineering Takeaways

- AI model vs. Application (Host) ➔ Orchestrator vs. Executor
- Applicaton acts as validator (AI model's output payload to be verified against JSON Schema)
- JSON RPC 2.0 (one standard protocol)